

Accurate Prediction of GPCR Ligand Binding Affinity with Free Energy Perturbation

Francesca Deflorian, Laura Perez-Benito, Eelke B Lenselink, Miles Congreve, Herman W. T. van Vlijmen, Jonathan S. Mason, Chris de Graaf,* and Gary Tresadern*



Cite This: *J. Chem. Inf. Model.* 2020, 60, 5563–5579



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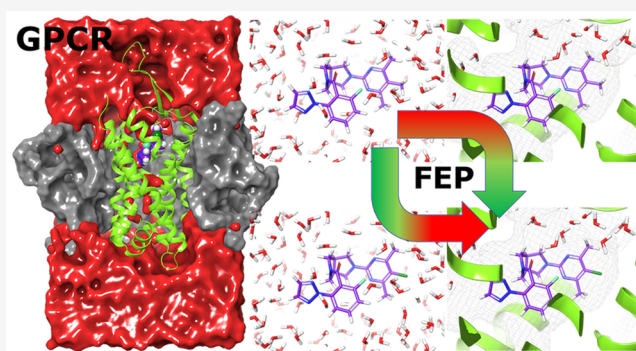


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ABSTRACT: The computational prediction of relative binding free energies is a crucial goal for drug discovery, and G protein-coupled receptors (GPCRs) are arguably the most important drug target class. However, they present increased complexity to model compared to soluble globular proteins. Despite breakthroughs, experimental X-ray crystal and cryo-EM structures are challenging to attain, meaning computational models of the receptor and ligand binding mode are sometimes necessary. This leads to uncertainty in understanding ligand–protein binding induced changes such as, water positioning and displacement, side chain positioning, hydrogen bond networks, and the overall structure of the hydration shell around the ligand and protein. In other words, the very elements that define structure activity relationships (SARs) and are crucial for accurate binding free energy calculations are typically more uncertain for GPCRs. In this work we use free energy perturbation (FEP) to predict the relative binding free energies for ligands of two different GPCRs. We pinpoint the key aspects for success such as the important role of key water molecules, amino acid ionization states, and the benefit of equilibration with specific ligands. Initial calculations following typical FEP setup and execution protocols delivered no correlation with experiment, but we show how results are improved in a logical and systematic way. This approach gave, in the best cases, a coefficient of determination (R^2) compared with experiment in the range of 0.6–0.9 and mean unsigned errors compared to experiment of 0.6–0.7 kcal/mol. We anticipate that our findings will be applicable to other difficult-to-model protein ligand data sets and be of wide interest for the community to continue improving FE binding energy predictions.



INTRODUCTION

Predicting relative protein ligand binding affinities is of high interest for drug discovery.^{1–3} Free-energy perturbation (FEP)⁴ is a class of relative binding free energy (RBFE) calculation based on a rigorous approach that typically uses molecular dynamics (MD) simulations to compute the binding free-energy difference between two structurally related ligands.⁵ The earliest applications of RBFE date back several decades,^{6–10} but it is experiencing a resurgence due to improved force fields, hardware improvements, and improved ease of use.^{11–13} Modern implementations often exploit highly parallelized GPU hardware architecture, such as FEP+ from Schrödinger¹⁴ that uses the Desmond MD-engine.¹⁵ There have been many recent valuable applications^{16–18} including work from our laboratories^{19–21} reporting mean unsigned error (MUE) between calculation and experiment <1 kcal/mol. Thus, RBFE calculations are becoming a widely used tool for computational drug discovery, but success is far from guaranteed. Improvements to force fields is an area receiving attention,^{22,23} but other key aspects include the initial system setup and sampling.²⁴ A recent study suggested high quality

experimental input structures were a requirement for success.²⁵ Meanwhile, sampling issues such as slow loop motions around the binding site,²⁶ larger protein movements,²⁷ insertion of buried waters, or unexpected ligand conformational changes²⁸ can be challenging for alchemical RBFE methods.

G protein-coupled receptors (GPCRs) represent the largest single target family for marketed drugs,²⁹ with growing opportunities for structure-based drug discovery,³⁰ and therefore various reported GPCR RBFE applications. In 2005, the binding energy difference ($\Delta\Delta G$) for several amino acid mutants of the cholecystokinin-9 (CCK9) peptide agonist binding to the CCK1 receptor delivered absolute errors <0.8 kcal/mol compared to experiment.³¹ Subsequently, Keränen et al. used alchemical transformations to predict the effect of

Special Issue: Novel Directions in Free Energy Methods and Applications

Received: April 29, 2020

Published: June 15, 2020



ACS Publications

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5563

<https://dx.doi.org/10.1021/acs.jcim.0c00449>
J. Chem. Inf. Model. 2020, 60, 5563–5579

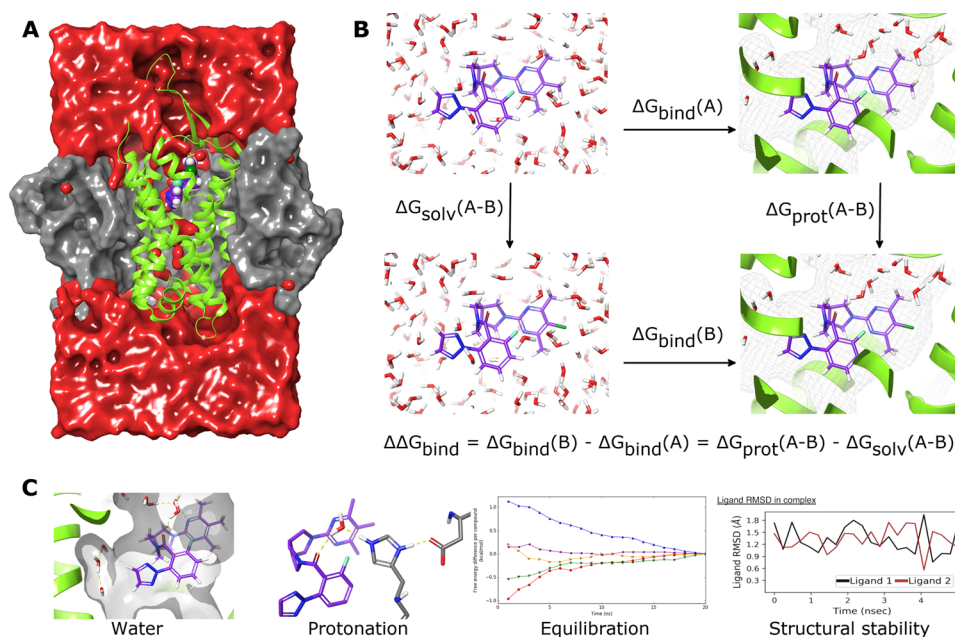


Figure 1. Overview of the approach. (A) Typical system comprising the transmembrane GPCR (green), the lipid membrane (gray), solvent (red), and ligand (magenta). (B) FEP relative binding free energy thermodynamic cycle showing how the difference in the binding energy of molecules A and B ($\Delta\Delta G_{\text{bind}}$) can be determined from the difference in energies of the perturbation of A to B in both solvent and in protein, $\Delta G_{\text{solv}}(A-B)$ and $\Delta G_{\text{prot}}(A-B)$ respectively. (C) Schematic view of the themes that have been studied in this report.

amino acid mutations on ligands ZM241385 and NECA binding to the Adenosine 2A (A_{2A}) receptor.^{32,33} Quantitative assessment of the MUE was complicated due to inactive mutant-ligand combinations, but trends were encouraging, and the calculations showed the crucial role of solvent-mediated interactions in ligand binding. More recently, a similar approach was used to study mutations at Y1 and Y2 neuropeptide receptors.^{34,35} Lee et al. used FEP to calculate absolute binding free energies of an inverse agonist, a neutral antagonist, and an agonist to β_2 -adrenergic receptor (β_2 -AR) active and inactive states, respectively³⁶ but the MUE was >2 kcal/mol. In 2015, RBFs were calculated for four kappa opioid receptor antagonists³⁷ although experimental data was only available for one of the perturbations. The largest study to date was published in 2016 by Lenselink et al.³⁸ In retrospective calculations, the relative binding energy of 45 ligands for five GPCR data sets gave a favorable average MUE of <1 kcal/mol for the ΔG 's; however, 3 of the 5 data sets showed low correlation, $R^2 < 0.3$. In their prospective study, four compounds were synthesized for the A_{2A} receptor and two had MUE < 1 kcal/mol whereas the other two were overpredicted by >2.5 kcal/mol compared to experiment, due to the protonation state of a histidine close to the ligand binding site. Cappel et al. performed FEP using 9 ligands and an A_{2A} receptor homology model,³⁹ and results gave R^2 of 0.47 but a high MUE of 1.50 kcal/mol. Jespers et al. used the Q- software program to calculate FEP selectivity for ligands binding at multiple adenosine receptors.^{40,41} Data were reported for three perturbations, with two having average MUE ~ 0.9 kcal/mol. Finally, Matricon et al. also used FEP to study ligand binding at the A_{2A} receptor and reported $R^2 = 0.75$ for $\Delta\Delta G$.⁴² Overall, although there are various publications of RBFE calculations applied to GPCRs, closer inspection reveals the results to be less convincing than for soluble proteins for which there are dozens of reports and several large-scale studies.^{14,43} In contrast, GPCR RBFE studies tend to be

smaller and rarely combine good R^2 and MUE. This can be due to small data sets, and small dynamic range in bioactivity that leads to relatively low errors but poor correlation due to predictions being clustered around the mean. As we will demonstrate, the application to this class of membrane bound receptors presents substantial difficulties, but sensible strategies can be developed, and robust results achieved.

The GPCR field, and especially the computational study of GPCRs, has undergone a revolution in the last 10 years with the breakthroughs in structural biology. Along with others, research at Sosei Heptares has played a major role in this new era of structure enabled GPCR molecular design.⁴⁴ With a wealth of unexpected binding sites and modes, modeling without reliance on at least a very close X-ray crystal structure can be problematic. For instance, traditional pharmacophoric or conformational superposition have been shown to be insufficient to predict the binding modes in the case of both glutamate⁴⁵ and orexin receptors.⁴⁶ Furthermore, GPCR structure activity relationships (SAR) are often driven by lipophilic hotspots, water hydrogen bond networks, key but scarce polar interactions that are sometimes water bridged, and the displacement of energetically unfavorable water molecules.^{47,48} Likewise, small detailed changes in the binding site such as compression or opening of the extra-cellular facing region, or side-chain dihedral rotations are hallmarks of how ligands elicit functional effects. In the absence of high-resolution structures many of these details are overlooked or even modeled incorrectly. Short-time scale MD simulations are usually insufficient to sample these effects, or incapable of correcting problems, and therefore poor results arise. Popular drug target classes such as kinases, proteases, or phosphodiesterases can achieve high resolution experimental structures that address some of these concerns, but other target families can also suffer from similar uncertainties as those described above.

Traditionally GPCR MD modeling is seen as an expert-user scenario with a more complex input system comprising

Table 1. FEP and MM-GBSA Prediction of Binding Affinity of Triazine A_{2A} Antagonists^a

FEP-study	Dataset	Protocol	N cpds	λ (ns)	R ^{2b}	MUE ^c	RMSE ^c	Figure ^d
1	A _{2A} set 1	(HID278 ^{7,43}) Standard	3	20	0.38	1.51±0.40	1.65±0.43	2
2	A _{2A} set 1	(HID278 ^{7,43}) Standard + GCMC	3	15	0.78	1.16±0.25	1.24±0.35	2
3	A _{2A} set 1	(HIP278 ^{7,43}) Standard + GCMC	3	20	0.77	1.80±0.38	1.92±0.56	-
3b	A _{2A} set 1	(HIP278 ^{7,43}) Standard + GCMC	3	40	0.62	1.34±0.48	1.33±0.48	-
4	A _{2A} set 2	(HID278 ^{7,43}) Standard + GCMC	12	20	0.34	1.29±0.21	1.58±0.21	3G
5	A _{2A} set 3 (phenol) ^e	(HIP278 ^{7,43}) Standard + GCMC	4	40	0.56	1.37±0.50	1.66±0.30	3H
6	A _{2A} set 4 (phenyl)	(HIP278 ^{7,43}) Standard + GCMC	4	40	0.93	0.60±0.23	0.80±0.21	3I
7	A _{2A} set 5 (pyridine)	(HIP278 ^{7,43}) Standard + GCMC	4	40	0.98	1.04±0.21	1.14±0.16	3J
8	A _{2A} set 6 (pyridine)	(HIP278 ^{7,43}) Standard + GCMC	10	40	0.90	0.84±0.15	1.00±0.14	3K
9	A _{2A} set 6 (pyridine)	(HIP278) Standard + GCMC + extended system equilibration	10	20	0.89	0.81±0.15	0.98±0.14	3L
-	A _{2A} set 6 (pyridine)	MM-GBSA – default	10	-	0.54	1.93±0.63	2.69±0.90	-
-		MM-GBSA – 10 Å flexibility	10	-	0.20	5.46±1.00	6.23±2.07	-

^aChemical structures of each compound, the pK_i (the number below structure), protocols of each FEP-study, and performance compared to experiment (R², MUE, RMSE). ^bThe R² is coefficient calculated between ΔG calculated vs experiment. ^cMUE and RMSE are given in kilocalories per mole and were derived from the calculated ΔΔG's and all ± standard error. ^dThe figure showing the corresponding correlation plot of calculated vs experimental ΔG. ^eThe dichlorophenol of set 3 has no available experimental binding ΔG.

protein, lipids, ligand, solvent, and counterions requiring a bespoke treatment when compared to nonmembrane proteins.⁴⁹ This contrasts with modern-day RBEF calculations that use automated setup, equilibration, and production protocols that should work appropriately and without artifacts across all the many compounds under study. Bringing these aspects together we pose the questions: how does FEP perform on data sets of GPCR ligands relevant for drug discovery? Can standard protocols that are currently used “out-of-the-box” be applied to GPCRs? If not, how can performance be improved, and can we account for the key factors of GPCR structure-based design? Here we report FEP studies on two different class A GPCRs, the A_{2A} receptor and the orexin-2 (OX2) receptor.

The adenosine A_{2A} receptor is currently a target in immuno-oncology⁵⁰ and Parkinson's disease.⁵¹ A_{2A} is the first GPCR to be investigated clinically in immuno-oncology and the A_{2A} antagonist AZD4635/HTL1071 has potential use as a monotherapy or in combination with an anti-PD1 antibody for the treatment of cancer.⁵² AZD4635/HTL1071 and the triazine A_{2A} antagonist data set used to develop and evaluate A_{2A} FEP models in the current work (Table 1) have been identified and optimized using structure-based drug design.⁵³ Orexin receptors are currently targets in the treatment of sleep disorders and preclinical studies have concluded that antagonizing the OX receptors is efficacious in promoting sleep.^{54,55} The OX2 receptor antagonist series used in the study led to the identification of clinical candidate JNJ-

Table 2. FEP and MM-GBSA Prediction of Binding Affinity of Pyrrolopyrrole OX2 Antagonists^c

FEP-study	Dataset	Protocol	N cpds	λ (ns)	R^2 ^a	MUE ^b	RMSE ^b	Figure ^c
10	OX ₂ - 1	Standard	27	5	0.02	1.37±0.17	1.77±0.20	4B
11	OX ₂ - 1	Standard+GCMC	27	5	0.17	1.43±0.15	1.73±0.20	S2
12	OX ₂ - 1	Standard+WaterMap	27	5	0.20	1.16±0.14	1.48±0.16	S3
13	OX ₂ - 1	Rebuild	27	5	0.65	0.84±0.09	1.05±0.13	4D
14	OX ₂ - 1	Rebuild+GCMC	27	5	0.47	1.08±0.14	1.33±0.16	S4
15	OX ₂ - 1	Rebuild	27	1ns x5 ^d	0.60	0.94±0.10	1.15±0.14	S5
16	OX ₂ - 1	Rebuild+HID350 ^{7,39}	27	5	0.33	1.05±0.11	1.25±0.12	S6
-	OX ₂ - 1	MM-GBSA – default	27	-	0.35	3.08±0.54	4.14±0.81	-
-	OX ₂ - 1	MM-GBSA – 10 Å flexibility	27	-	0.11	4.02±0.55	4.90±0.96	-

OX ₂ set 2: 24 mols	1 8.1	1 8.0	1 8.0	1 7.9	1 7.9	1 7.7	1 7.7	1 7.7	1 7.3	1 7.3	1 7.3	1 7.1	1 7.1	1 5.0
2 7.1	2 7.1	2 7.0	2 6.9	2 6.9	2 6.9	2 6.6	2 6.3	2 6.3	2 5.8	2 5.8	2 5.0	2 5.0	2 5.0	2 5.0

OX ₂ - 2	Standard	24	5	0.06	1.33±0.14	1.61±0.18	S7
OX ₂ - 2	Rebuild	24	5	0.47	0.80±0.09	0.98±0.09	5B
OX ₂ - 2	Rebuild+equi-with-set02 compound	24	5	0.62	0.67±0.08	0.84±0.11	5C
OX ₂ - 2	MM-GBSA – default	24	-	0.10	5.01±0.83	6.40±1.34	-
OX ₂ - 2	MM-GBSA – 10 Å flexibility	24	-	0.03	4.25±0.65	5.27±1.05	-

^aThe R^2 coefficient is calculated between ΔG calculated vs experiment. ^bMUE and RMSE are given in kilocalories per mole and were derived from the calculated $\Delta\Delta G$'s and all $\pm$ standard error. ^cThe figure showing the corresponding correlation plot of calculated vs experimental ΔG . ^dFive repeats of 1 ns, results averaged. ^eChemical structures of the datasets studied with pKi of each molecule, protocols applied in each FEP-study, and performance compared to experiment (R^2 , MUE, RMSE).

42847922.^{56–58} The two A_{2A} and OX₂ receptors illustrate the potential and challenges of RBFE calculations for GPCRs and the need for customized settings. Naturally, both receptors are membrane bound with binding sites deep in the receptor and ligand binding that involves trapping of key waters. The A_{2A} data set includes X-ray complexes solved with a triazine antagonist series (Table 1),^{43,59} and we investigated the simulation time, the protonation state of histidines (and the effect on salt bridges) and water networks (enabling trapped

water environments to change during the simulations). The latter aspect was critical for success in all studies, indeed, for the A_{2A} triazine set having an occluded site with missing waters led to a complete failure to predict binding affinity. The OX₂ example has no available X-ray structure with the ligands used (Table 2), but there are other high resolution structures of similar ligands.⁴⁶ The data set mimics the common scenario to setup and test aspects of the binding site and mode: such as the protein, ligand position, and water molecule positions. We

show how detailed consideration of ligand and receptor stability during simulations, side-chain ionization states, and water placement all led to substantial improvements that would be suitable for FEP guided drug discovery, Figure 1 provides an overview of the study. The average MUE on the predicted $\Delta\Delta G$ for the best FEP protocols were 0.60 ± 0.23 and 0.67 ± 0.08 kcal/mol, and importantly R^2 was also good, 0.93 and 0.62, for the best cases of A_{2A} and OX2 respectively.

METHODS

Selecting Data Sets. Antagonists of the A_{2A} receptor were selected from the Sosei Heptares work on 1,2,4-triazine derivatives.⁵³ A first data set with 12 triazine derivatives was chosen with modifications located on ring A (Table 1) with equal representatives of a pyridine, phenyl, and phenol ring derivatives. The pKi values of this training set of triazines span from 6.5 to 8.8. First tests were performed with 3 examples from this set, then using all 12 examples. An additional set of 6 variously decorated but congeneric pyridine-triazine compounds was used to validate the setup and parameters of the FEP calculations and test their predictive power for a new set of compounds.

Two data sets of octahydropyrrolo[3,4-*c*]pyrrole OX2 antagonists were compiled from previous Janssen reports.^{56,60} The compounds featured modification in two distinct areas but are based on the same chemical scaffold (Table 2). The first set of 27 compounds had a range of pKi from 5.0 to 8.5, while the second set comprised 24 compounds and pKi ranged from 5.0 to 8.1. Both sets are representative of typical lead optimization (LO) structural modifications but at the same time are considered in the scope of relative alchemical FEP calculations, involving modifications of substituents on the same pyrimidine or phenyl ring, respectively. As in the case of A_{2A} , the second set was intended as an external validation of the FEP model derived for set 1. Table S1 provides the full chemical structures and bioactivities of the ligands in both data sets. The compound activity for both GPCRs was measured using radioligand binding displacement assays, pKi, preferable to functional assays for which it is difficult to decouple affinity and functional effect. The pKi were converted to expected binding free energies at 298 K using the relation $RT \ln(K_i)$, any offset with respect to binding free energy cancels when comparing relative differences.⁶¹

System Preparation. For A_{2A} the high-resolution crystallographic structures in complex with triazines HTL929, compound 4e in this work, (PDB ID SOLZ⁵⁵) and HTL1071 (PDB ID 6GT3) were used to create the FEP input systems. Prime⁶² was used to back-mutate the stabilizing mutations of the crystallized receptor and to build the intracellular loop 3 (between Leu208 and Glu219) after removal of the fusion protein. The water molecules and the sodium ion from the high-resolution structures were included.

The system-builder in Maestro was used to prepare the simulation inputs, a membrane of pre-equilibrated (300 K) POPC lipids based on the OPM⁶³ database alignment, SPC solvent with an orthorhombic box and the OPLS3e force field. Unless otherwise stated, all protein structures were prepared using the "Protein Preparation Wizard"⁶⁴ tool in Maestro with default settings, missing side chains and atoms were fixed, protein protonation states were assigned with PROPKA at pH 7.0, zero bond order constraints to neighboring atoms were assigned, the hydrogen bonding network was optimized, ligand charges assigned, and a brief minimization to a root-mean-

square deviation (RMSD) of up to 0.5 Å was performed. For A_{2A} two different protonation states of the residue H278^{7,43} were considered: the neutral form HID (N δ 1-protonated π tautomer) and the positively charged form HIP (cationic imidazolium). Equilibration followed a seven step routine: (i) 50 ps at temperature of 10 K with Brownian dynamics, NVT ensemble, solute heavy atom restraints and small (1 fs) time step; (ii) 20 ps at 100 K with Brownian dynamics, NPT ensemble, high pressure, membrane and protein restraints, and 2 fs time step; (iii) 100 ps at 100 K with MTK barostat, NPgT ensemble, membrane and protein restraints, and 2 fs time step; (iv) 150 ps heating from 100 to 300 K with MTK barostat, NPgT ensemble, lower membrane and protein restraints, and 2 fs time step; (v) 50 ps at 300 K with Nosé–Hoover barostat, NVT ensemble and restraints on protein backbone and ligand, timesteps increased to defaults; (vi) 50 ps at 300 K with Nosé–Hoover barostat, NVT ensemble and no restraints; (vii) 100 ps at 300 K with MTK barostat and NPgT ensemble and no restraints.

For OX2 the PDB structure 4S0V⁶⁵ was used. This structure was chosen as it is bound to suvorexant, a ligand with high structural similarity to data sets 1 and 2. The fusion protein was removed, and the loop closed with three glycine residues linking Q205 to K209 at the bottom of transmembrane (TM) helices 5 and 6 respectively. The extracellular loop 2 (ECL2) was not fully resolved and was modeled based on the OX1 structure PDB 4ZJ8.⁶⁶ Ligand alignment for all simulations was based on superposition with suvorexant, subsequent molecules from each data set were aligned using flexible ligand superposition available in Maestro, with manual modifications to ensure consistent alignment of the substituents on the pyrimidine ring in data set 1. In the standard protocol waters were retained from the crystal structure, further calculations used waters taken from other OX2 PDB structures, 5WQC and 5WS3.⁶⁷ As for A_{2A} , alternative HID and HIP tautomeric and protomeric forms of histidine were investigated. For some calculations the WaterMap⁶⁸ tool was used with default settings to identify stable water sites in the binding site that could then be included in the system input. The remaining protein preparation and equilibration protocols were the same as mentioned above for A_{2A} .

FEP+ Approach. FEP+ calculations were performed using v2019-1 and v2019-2 of the Schrödinger modeling suite. The OPLS3e force field, the Desmond (MD) engine v3.8.5.19, the replica exchange with solute tempering (REST-2),⁶⁹ and the multistate Bennett acceptance ratio (MBAR) approach to generate free energy estimates were used.⁷⁰ The REST region was applied only to heavy atoms of the ligand. Any missing force field parameters were added by additional quantum mechanical calculations and fitting using the ffbuilder module. The FEP+ calculations were performed with 12 λ -windows and varying length (1, 5, 20, 40 ns) production simulations per window. In addition to equilibration of the initial system that was mentioned above, prior to each λ -window further equilibration was performed in five steps: (i) 100 ps at temperature of 10 K with Brownian dynamics, NVT ensemble, solute heavy atom restraints, and small (1 fs) time step; (ii) 12 ps at 10 K with Berendsen thermostat, NVT ensemble, solute heavy atom restraints, and small time step; (iii) 12 ps at 10 K with Berendsen, NPT ensemble, solute heavy atom restraints, increase to default timesteps; (iv) 24 ps at 300 K with Berendsen, NPT ensemble, solute heavy atom restraints; (v) finally, 240 ps at 300 K with Berendsen, NPT ensemble and no

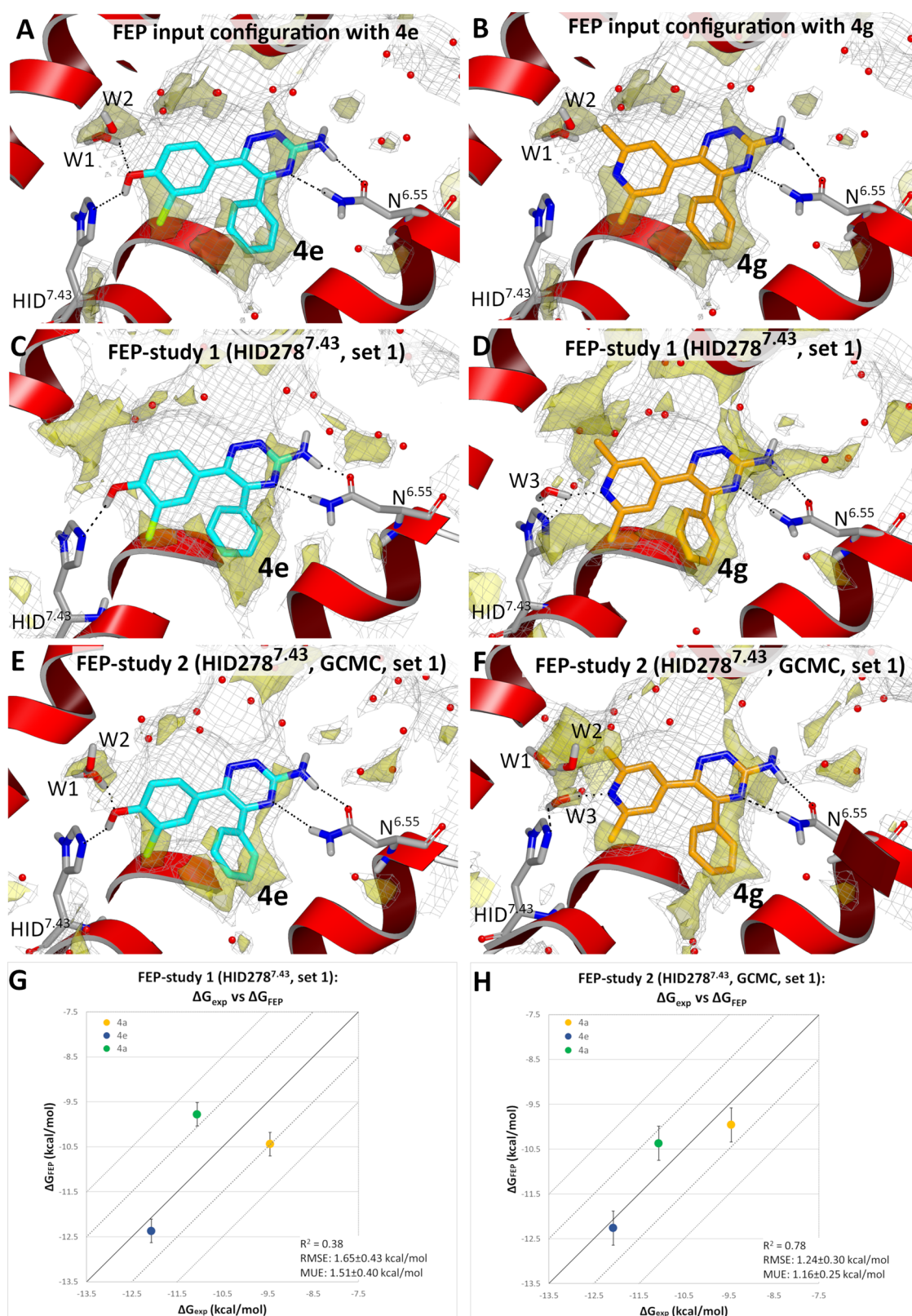


Figure 2. A_{2A} binding site and FEP results for FEP-studies 1 and 2: Comparison of input configurations with triazines 4e (A) and 4g (B) versus representative snapshots from MD simulations for FEP-study 1 without GCMC solvent sampling (C and D) and FEP-study 2 with GCMC solvent sampling (E and F). GRID maps shown as surfaces: CH3 probe in gray mesh at 1 kcal/mol defines the pocket surface in terms of how close a ligand carbon atom can go, and sp² CH probe (C1 =) in yellow transparent solid at −2.8 kcal/mol defines lipophilic hotspot subpockets. The use of the GCMC stage leads to stable hydration sites w1 and w2 for 4e (E) and also the extra hydration site w3 for 4g (F) bridging between His278^{7.43} and the pyridine ring. Only polar hydrogen atoms involved in A_{2A}-ligand hydrogen bond interactions and hydrogen atoms of water molecules in conserved locations forming a H278^{7.43} mediated hydrogen bond network with 4e and 4g (w1, w2, w3) are shown. FEP results for A_{2A} FEP-studies 1 (G) and 2 (H) are shown with the correlation of calculated vs experimental ΔG (kcal/mol).

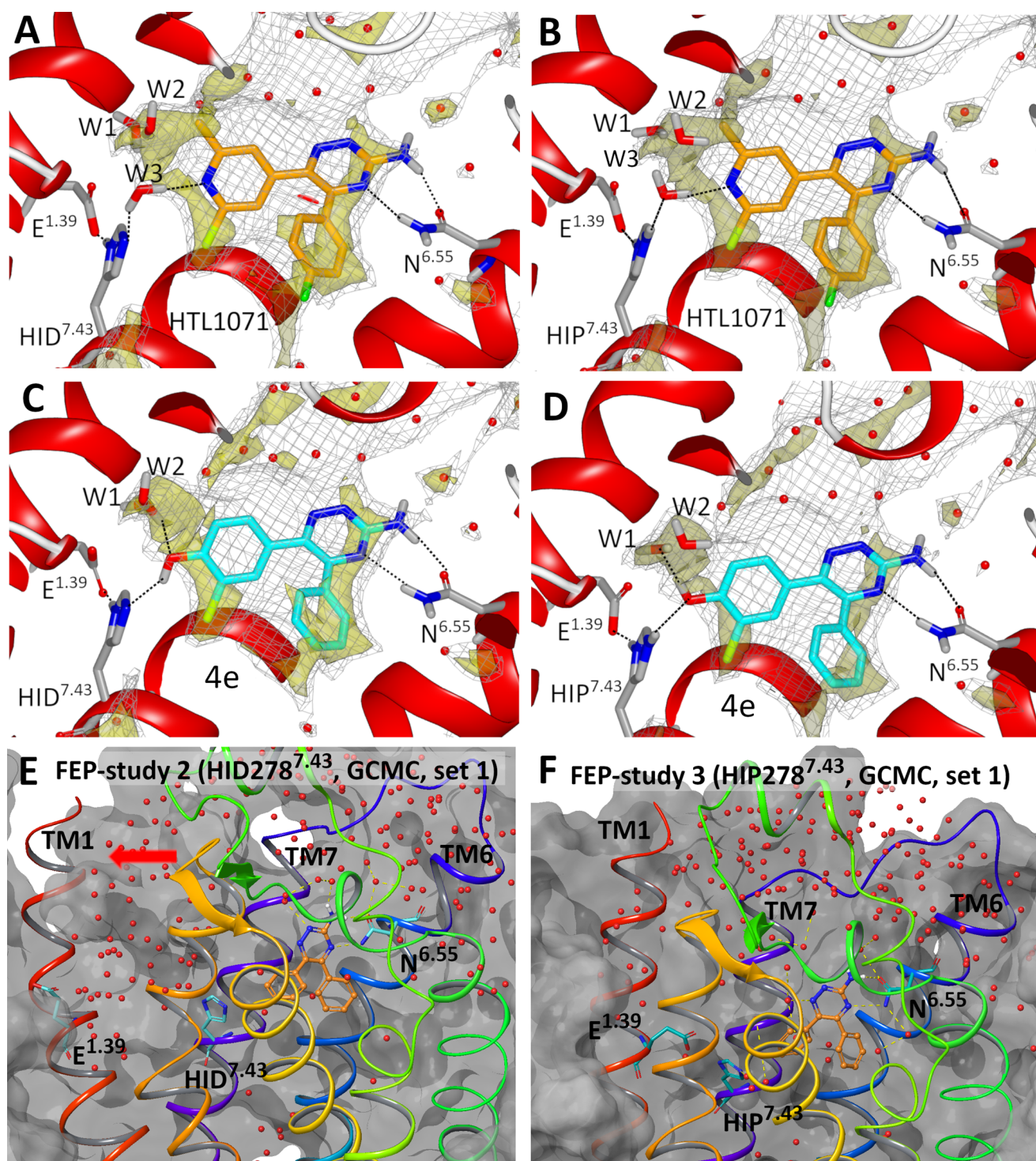


Figure 3. continued

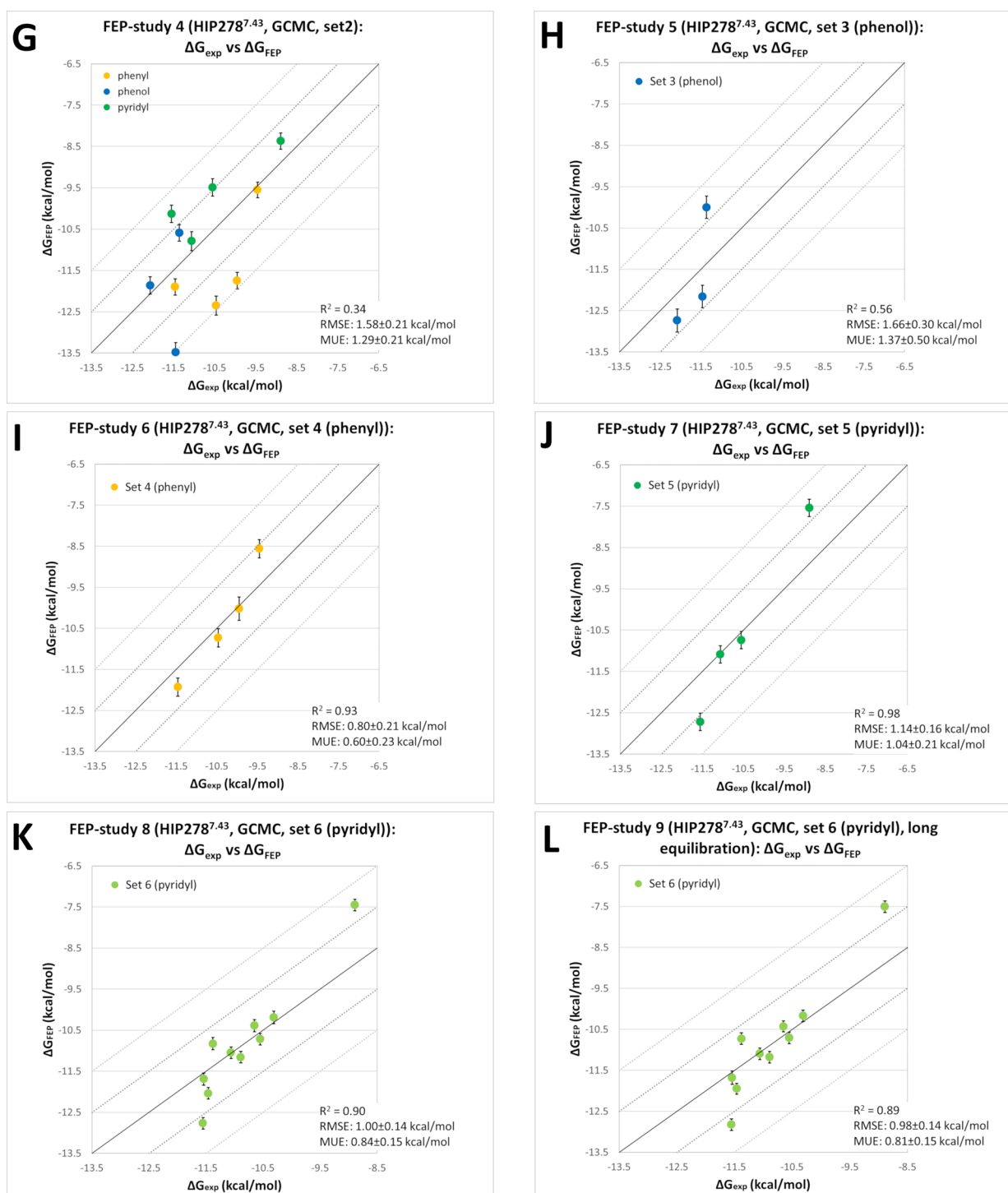


Figure 3. H278^{7.43} protonation states and water networks in binding sites of high-resolution crystallographic structures with pyridine–triazine HTL1071 (A, B) and 4e (C, D) in A_{2A} receptor FEP-studies 4–9. The protonation state of H278^{7.43} is considered in neutral form (HID278^{7.43}) in FEP-study 4 (A, C, E, G) and positively ionized (HIP278^{7.43}) in FEP-studies 5–9 (B, D, F, H–L). Only polar hydrogen atoms involved in A_{2A}–ligand hydrogen bond interactions and hydrogen atoms of water molecules in conserved locations forming a H278^{7.43} mediated hydrogen bond network with HTL1071 and 4e (w1, w2, w3) are shown. Rendering and color coding of binding site surface (gray mesh) and lipophilic regions (solid yellow) using GRID are as defined for Figure 2. Comparison of representative snapshots from MD simulations for FEP-study 2 (E) and FEP-study 3 (F) showing the higher stability of the system during MD simulations. An outward movement of the extracellular part of TM1 was observed during MD simulations for the system built with neutral HID278^{7.43} state. This movement allowed water molecules to enter the binding site disrupting the hydrogen bond interaction with E13^{1.39} (E), while the stronger charge–charge interaction between HIP278^{7.43} and E13^{1.39} stabilized the receptor (F).

restraints. Where indicated, some calculations used presolvation that was included via two additional 500 ps grand canonical Monte Carlo (GCMC) simulations to solvate the

pocket and the entire system performed at 300 K.^{71,72} It is important to clarify the differences in the two solvation methods, GCMC and WaterMap. The GCMC approach is

used to accelerate the sampling of water molecules. The GCMC simulation is performed at constant chemical potential, volume, and temperature, thereby allowing for the insertion and deletion of water in and out of the system. This is performed in an automated fashion at the start of an FEP simulation to add water around the ligands and to equilibrate the water density in the whole simulation volume. It is used to enable improved sampling of water positions and occupancy than can be achieved by MD alone. Thus, it can result in the introduction of different waters that fill voids or cavities for specific ligand perturbations. Meanwhile, WaterMap is an MD simulation performed on the protein ligand complex that identifies water sites around the ligand in the binding site, including their energetic stability. It is performed independently from, and prior to, the FEP calculations and the analysis leads to the identification of specific water molecules that will be included in all subsequent FEP calculations. WaterMap has been shown to perform well at recapitulating experimental water positions and is therefore useful for predicting solvation in the binding sites of low-resolution structures. In order to optimize and shorten the production time of the A_{2A} FEP+ protocol, the equilibration steps were extended by 10-fold and with the GCMC simulations extended by 3-fold. Production simulations used the NPT ensemble and hydrogen mass repartitioning to permit a 4 fs time step.

Analysis. The parameters and tools mentioned above were combined in different run-protocols for the FEP calculations, referred to as FEP-studies. The resulting calculated $\Delta\Delta G$'s were fitted to the mean of the experimental ΔG to deliver the calculated ΔG and reduce offset. Comparison with experiment was then performed via the coefficient of determination (R^2) of ΔG , and the MUE and RMSE, with the latter two calculated from the actual $\Delta\Delta G$'s. Error bars for ΔG_{FEP} in the correlation plots with experiment are derived from the cycle closure error for ΔG prediction across the many paths in the perturbation network.¹⁴ Experimental results were also compared to clogP (calculated with BioByte toolkit v2016) to provide an estimate of the correlation with experiment that can be expected from a baseline model and, indeed, the underlying ease with which correlation can be attained. Additional performance comparison was performed with molecular mechanics generalized Born surface area (MM-GBSA) calculations. These calculations were run using the specific inputs from certain FEP studies. An implicit variable dielectric surface generalized Born (VSGB) solvation model was used along with the OPLS3e force field. Two protocols were applied involving a default approach with no protein minimization and a second approach with minimization of a 10 Å radius of the surrounding binding site (using the same active region for all ligands). For fair comparison, the calculated MM-GBSA ΔG were also fit to the mean of the experimental ΔG . We also calculated the expected correlation coefficients⁷³ and then R^2 by sampling the experimental ΔG data 10000 times with a mean random error, then deriving the overall average correlation coefficient with the original experimental data. Error estimates of 0.3 and 1.0 log units were used to represent approximate experimental and FEP errors, respectively. Finally, binding site structures were analyzed and visualized using GRID⁷⁴ analysis. The choice of probes, energy contour levels, etc., was applied according to previously optimized settings for GPCR binding sites,^{48,75} showing the energetically favorable regions for the C1 = lipophilic probe (−2.8 kcal/mol) and CH3 probe (1 kcal/mol) defining the pocket surface.

RESULTS

A_{2A} FEP+ Results. The $\Delta\Delta G$ for the A_{2A} ligands was calculated with multiple protocols to evaluate different aspects of the FEP predictions including ligand dependent binding site water solvation with the use of a GCMC solvation step, different protonation states of binding site residue H274^{7,43}, and the optimization of FEP simulation equilibration vs production run simulation time, Table 1. Before discussing the FEP results, we first report the R^2 coefficient 0.17 and MUE 0.93 ± 0.78 kcal/mol between the binding affinity and a simple prediction based on clogP using all the triazine derivatives in Table 1. It has long been understood that even very simple descriptors can provide enrichment over random performance⁷⁶ and their application reveals the ease with which a data set can be separated into actives and inactives or ranked by activity. In this case, clogP reveals very little baseline correlation but a relatively small error in prediction due to the narrow range of predicted logP, with mean clogP of 2.8 ± 0.60 log units. The large standard deviation in the MUE (± 0.78 kcal/mol) indicates that there was a large spread in the accuracy of the individual predictions. Initial FEP tests used a small training set of 3 representative triazines from set 1: 4a without polar substituent, pyridine 4g and phenol 4e, caution is needed to not over interpret the significance of R^2 for such small data sets. Each requires a different water mediated hydrogen bond interaction network⁷⁷ (Table 1) therefore providing a test of GCMC water placement simulations. In this regard, GCMC predicted the ligand dependent A_{2A} binding site solvation consistent with high resolution crystal structures, leading to the correct ranking of 4a, 4g, and 4e with FEP (Table 1 and Figure 2), FEP-study 2, R^2 and MUE 0.78 and 1.16 ± 0.25 kcal/mol respectively, compared to FEP-study 1 0.38 and 1.51 ± 0.40 kcal/mol. Without GCMC, no stable water networks were observed deep in the binding site near H278^{7,43}, resulting in overprediction of the affinity of 4a lacking polar substituents and the underprediction of the affinity of pyridine 4g, lacking the H278^{7,43} stabilized water molecule observed in the GCMC FEP simulations (Table 1 and Figure 2).

We subsequently extended the ligand set with further analogs of the three representatives to evaluate the accuracy of the FEP protocol in combination with GCMC to predict ligand affinities. The performance of FEP-study 4 was lower than expected from the experience with the small training set, leading to a negative outcome on the consistency and robustness of the FEP model with R^2 0.34 and MUE 1.29 ± 0.21 kcal/mol (Table 1 and Figure 3). This was substantially lower than the expected R^2 of 0.83 based on experimental error, but similar to an estimate of 1 log unit error in FEP, R^2 of 0.27, both calculated for the same set of 12 triazine compounds. Therefore, it was clear there was room for improvement of the FEP protocol.

We evaluated different protonation states of His278^{7,43} that has been postulated to be in an equilibrium between neutral histidine with δ -nitrogen (HID) protonated, and +1 charge histidine with both δ -nitrogen and ϵ -nitrogen protonated (HIP) (Figure 3). Using the neutral histidine, HID278^{7,39}, an outward movement of the extracellular TM1 was observed, allowing water to enter the A_{2A} binding site (Figure 3). Starting from the thermalized system of one complex and consideration of HIP278^{7,43} protonation state provided stability of the A_{2A} antagonist complexes during simulations (Figure 3) but did

not result in a more accurate prediction of $\Delta\Delta G$ for the small triazine set, FEP-study 3, R^2 0.77 and MUE 1.80 ± 0.38 kcal/mol (Table 1 and Figure 3) compared to MUE 1.16 ± 0.25 kcal/mol for FEP-study 2. Longer simulations of up to 40 ns per λ window of the same system and same protocol showed better convergence and achieved comparable MUE and RMSE values to the previous runs (FEP-study 3b, R^2 0.62 and MUE 1.34 ± 0.48 kcal/mol).

Comparative analysis of the FEP simulations indicated that the flexibility of the A_{2A} binding site and ligand chemotype dependent binding site solvation required the use of representative initial and ligand reference structures to define independent FEP models for the different subseries (Table 1 and Figure 3). This ligand customized approach and the consideration of long 40 ns simulations giving improved convergence, led to the formation of optimal receptor–ligand interactions and shape complementarity during FEP. Likewise, the hydrogen bond network in the buried pocket was better sampled and stabilized. For example, FEP-studies 6 and 7 built from just subseries of phenyl (4a, 4b, 4c, and 4d) and pyridyl (4g, 930, 1037, and 1056) compounds, showed improved R^2 0.93 and 0.98 and MUE 0.60 ± 0.23 and 1.04 ± 0.21 kcal/mol, respectively. Thermalizing the system with bulky substitutions further improved performance (R^2 0.54 and MUE 0.95 ± 0.22 kcal/mol), resulting in FEP models that allowed accurate ranking of different analogues and provided a robust model for the set using all ten pyridine examples in FEP-study 8, with R^2 0.90 and MUE 0.84 ± 0.15 kcal/mol (Figures 2 and 4). Extension of the thermalization step (each stage 10 $\times$ longer, except the GCMC step that was extended 3 $\times$, to 1.5 ns) allowed faster convergence without affecting FEP+ performance, providing an alternative to the long 40 ns production simulations to reach similar convergence and predictive quality, FEP-study 9 R^2 0.89 and MUE 0.81 ± 0.15 kcal/mol (Table 1). This was better than the expected R^2 of 0.76 and 0.19 based on experimental (0.3 log unit) and possible FEP (1 log unit) errors respectively, calculated for the same set of ten pyridyl-containing triazine compounds. Therefore, for this data set the optimized FEP model was overperforming compared with experiment, suggesting performance would decline when expanded to larger numbers of compounds. The MM-GBSA protocols both showed worse performance, with the best case R^2 being 0.54 and MUE of 1.93 ± 0.63 kcal/mol, further highlighting the improved performance of the optimal FEP studies.

OX2 Data Set 1 FEP+. We investigated multiple protocols to calculate the $\Delta\Delta G$ for the OX2 ligands in data set 1, Table 2. As for A_{2A} , we calculated the R^2 and MUE between the experimental binding energy and clogP, 0.17 and 1.53 ± 0.96 kcal/mol respectively for the set of 27 compounds. The results reveal no predictive ability for the simple model and in this case relatively large unsigned errors and with a large variance. The first FEP calculations delivered no correlation to experiment and large errors, R^2 0.02 and MUE 1.37 ± 0.17 kcal/mol FEP-study 1 Table 2. Augmenting the input with GCMC water placement or WaterMap stable water positions gave small improvements (FEP-studies 11 and 12, R^2 0.17 and 0.20, respectively). Close inspection of representative structures from the simulations revealed the ligand binding mode was opening up in a scissor-like motion and losing the compact geometry and the intramolecular interaction between the phenyl and pyrimidine rings, Figure S1. Indeed, the ligands did not remain stable during the MD simulations. For FEP-

study 10 using the standard setup, 22 of the total 40 perturbations involved ligand RMSDs exceeding 4 Å (a cutoff suggesting substantial ligand movement during an MD simulation). During the simulations the triazole moved deeper into the receptor pulling the central phenyl and distal pyrimidine rings apart, losing their intramolecular interaction and the ligand's horseshoe conformation. As the interaction was lost the distal pyrimidine, where the perturbation was taking place, drifted upward away from its initial binding subpocket. Furthermore, this was often accompanied by small opening movements of the TM helices also allowing the ligand(s) to adjust and move outward from the binding site.

We returned to the input to address these issues. Comparing with the OX2 crystal structure PDB id 5WQC⁶⁷ revealed two additional crystallographic waters that occupy the pocket below the triazine. The importance of these waters has recently been discussed for the binding mode of suvorexant.⁴⁶ It was apparent that their kinetic stability, being buried at the bottom of the pocket, would block the downward triazine movement that had been seen in the first FEP simulations. Closer inspection of the amino acid ionization states around the ligand binding site identified two histidine residues H224^{5,39} and H350^{7,39} on the extracellular side making direct interactions with corresponding acidic residues E212^{45,52} and D115^{2,65}. Assigning the correct tautomeric and protomeric state of histidines is notoriously difficult. Given their direct proximity to acidic side chains it was plausible they form a stronger charge–charge interaction to stabilize the receptor, again recently discussed for suvorexant,⁴⁶ and therefore, we protonated these two histidine side chains. Finally, WaterMap was used to define additional stable water sites around the ligand, as in FEP-study 12. The binding site of the original standard input is compared to the rebuilt input in Figure 4A and C. The differences arise due to small details such as the additional waters deep in the triazine pocket and additional ordered water structure above the ligand and hydrogen bonds around it. Interestingly, the use of protonated His residues would permit stronger interactions to the corresponding acidic side chains, but details of the water network such as the key water hydrogen bond to the ligand amide-carbonyl oxygen are still fulfilled, indeed the extra proton of the histidine is incorporated into the water interaction network with no disruption. Altogether it was expected that these changes would enhance the stability of the MD simulations, and likely better capture reality.

The modifications to the input, referred to as rebuild in Table 2, led to the desired improvements in ligand stability. Whereas previously 22 perturbations had involved ligand RMSD movements exceeding 4 Å, in this case only 2 of all the perturbations showed the same extent of ligand movement. This immediately translated to better affinity predictions between ligands and improved the overall correlation and error, R^2 0.65 and MUE of 0.84 ± 0.09 kcal/mol, Figure 4D and FEP-study 13 Table 2. The expected R^2 for the same 27 compounds were calculated to be 0.90 and 0.42 based on estimated experimental or FEP errors, respectively. Therefore, it is clear the calculated R^2 did not outperform experiment, and was performing better than the estimated FEP error. The FEP vs experimental ΔG correlation plot (Figure 4D) shows that some of the least active compounds are now better predicted than in FEP-study 10. Also, the $\Delta\Delta G$ error count in Figure 4D shows a more favorable distribution compared to the initial configuration in Figure 4B with more perturbations within 1

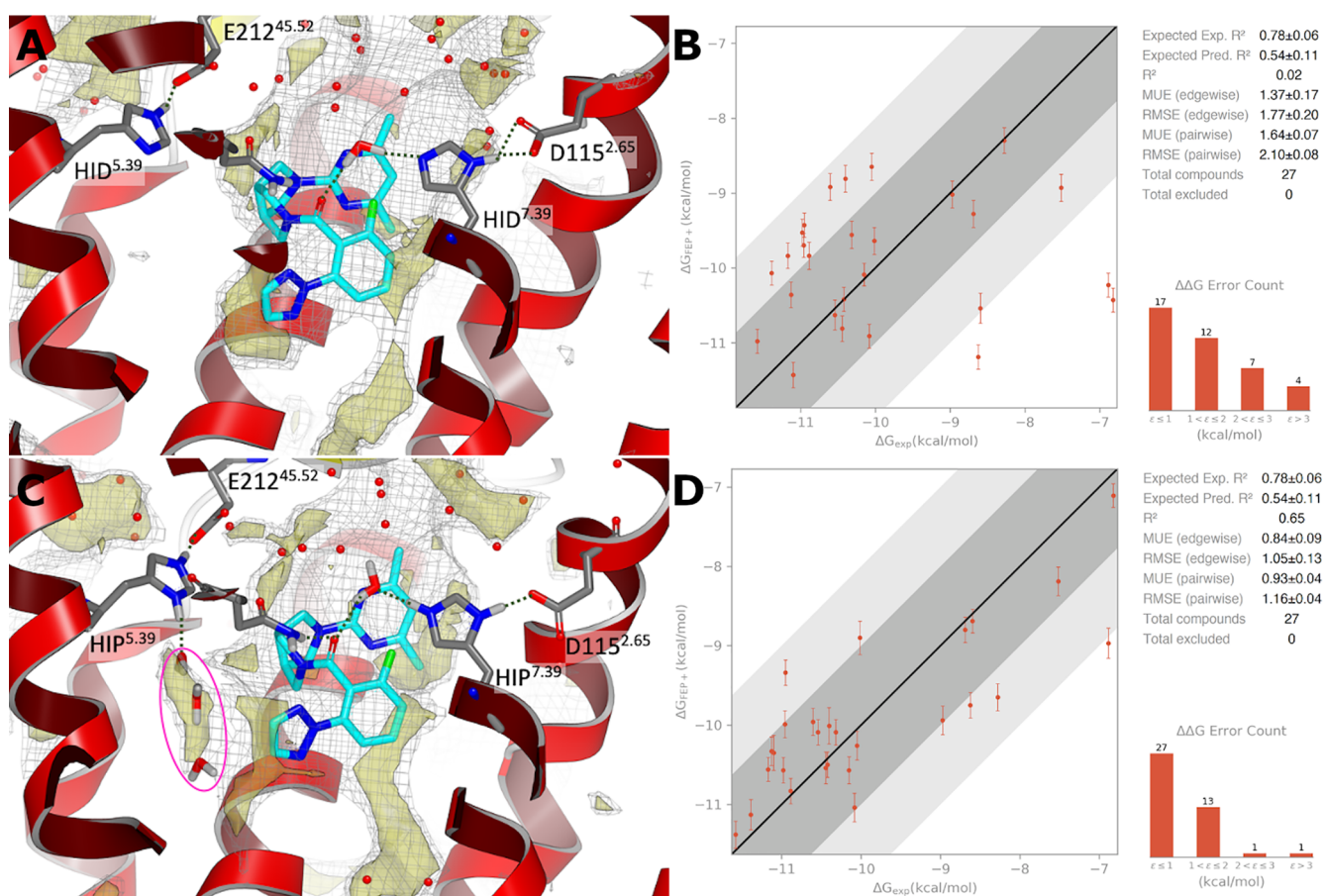


Figure 4. OX2 binding site and FEP results for data set 1. Comparison of input configurations and results using the standard setup protocol (A and B) versus the modified rebuilt system (C and D). The rebuilt system features extra crystallographic waters and a complete water network from the WaterMap output around and above the ligand. Histidine residues H224^{5.39} and H350^{7.39} were protonated and form direct salt bridge interactions with acidic amino acids E212^{45.52} and D115^{2.65}. The input structures (A and C) highlight the lipophilic hotspots in the OX2 receptor binding site, the rendering and color coding of binding site surface (gray mesh) and lipophilic regions (solid yellow) using GRID are as defined for Figure 2. FEP results (B and D) for OX2 data set 1 for each system are shown with the correlation of calculated vs experimental ΔG (kcal/mol). Performed using the protocol and input described in FEP-study 10 (B) and the rebuilt input in FEP-study 13 (D).

kcal/mol of experiment and fewer large errors >2 kcal/mol. It is well understood that ligands must be stable in FEP simulations to enhance adjacent λ window overlap and enable energy convergence. In addition, once the simulations are stable it becomes possible for FEP to capture the structure activity relationship (SAR). In this case the most active ligands must adopt and maintain the intramolecular π -stacked horseshoe conformation seen for suvorexant⁴² with the corresponding pyrimidine and aryl rings interacting with W120^{23.50} and H350^{7.39}, respectively. The pocket is mostly hydrophobic being lined by A110^{2.60}, T111^{2.61}, V114^{2.64}, I130^{3.28}, P131^{3.29}, and Y354^{7.43} while above the ligand there are three salt bridges stabilizing the extracellular side of the receptor, E118^{2.68}–R339^{7.28}, D211^{45.51}–R328^{6.59}, and E212^{45.52}–H224^{5.39}, as well as a water network bridging ligand and receptor.⁴² The SAR of modifications on the pyrimidine ring in set 1 is mostly driven by changes in the hydrophobic interactions. There are gains of potency for optimal small substituents that occupy the favorable lipophilic hotspot region visible in Figure 4C, such as the trimethyl substituted pyrimidine, pKi 8.4 Table 2. However, larger substituents do not fit and must rotate upward, where there is no favorable lipophilic hotspot, suggesting the water is relatively stable in this region. Examples such as the morpholine, furan, and

dimethylacetamide substituents are among the least active examples; thus their binding does not compensate for the energetic cost of disrupting the interactions and stable waters above the ligand.

Returning to the FEP results in Table 2, the use of the presolvation step with the GCMC simulation did not have a major impact on the results, FEP-study 14 Table 2. GCMC is particularly suited to insert buried waters where they otherwise would not enter in MD simulations. In this case the buried waters were already placed below the triazine as described above and were essential for ligand binding of all examples in this data set but were not displaced. Instead, the changes in solvation happening during ligand perturbation occur in the solvent accessible extracellular side of the binding site and were already captured in the standard approach.

It has been suggested that multiple repeats of free energy calculations can improve accuracy due to diminishing effects of random errors.⁷⁸ Here we tested this with five repeats of FEP simulations using 1 ns λ windows, FEP-study 15 Table 2. Correlation and MUE were just slightly worse, 0.60 and 0.94 ± 0.10 kcal/mol respectively compared to the single repeat using 5 ns simulations (FEP-study 13) despite a comparable total simulation time. Finally, we returned to the input structure and converted the protonated H350^{7.39} to the HID tautomeric

form that removes the formal charge but maintains the hydrogen bond donation to D115^{2.65}. No other changes were made to the input, but the calculations delivered worse performance, R^2 0.33 and MUE 1.05 ± 0.11 kcal/mol, FEP-study 16 Table 2. This suggests the positively charged His is beneficial for stability and convergence of the FEP MD simulations but may also be important for SAR by having an effect on protein–ligand interactions. Finally, for OX2 data set 1, the MM-GBSA calculations were performed using the analogous input to study 13 but delivered worse performance; the best case of R^2 was 0.35 and MUE 3.08 ± 0.54 kcal/mol.

OX2 Data Set 2 FEP+. To simulate a prospective validation scenario, we then turned to the second OX2 data set using it as an external test case. The compounds were from the same chemical series but involve perturbations on the aryl ring separating the pyrrolopyrrole scaffold and the distal 1,2,3-triazole, Table 2. Again, clogP delivered no predictive correlation for the experimental binding affinity, R^2 0.16. The MUE was relatively low, 1.13 ± 0.78 kcal/mol, as had been the case for the A_{2A} data set, highlighting the need for the quality of the prediction to be assessed in terms of both correlation and error. Turning to the FEP results, as had been the case with the first OX2 data set, using the standard prepared input delivered no correlation and poor MUE with experiment, R^2 0.06 and MUE 1.33 ± 0.14 kcal/mol FEP-study 17 Table 2. Again, a large proportion (11 out of 42) of perturbations showed unstable ligand RMSD exceeding 4 Å. Therefore, we immediately turned to the rebuilt input and expected good performance as seen with FEP-study 13 for data set 1. Although performance improved with respectable MUE of 0.80 ± 0.09 kcal/mol, FEP-study 18 Table 2, the overall correlation was still low, Figure 5B. In this case there were five perturbations that showed bad ligand RMSD values. Given the results above for A_{2A}, an immediate alternative approach was to equilibrate the system in the presence of a compound from data set; in this case the most potent compound was chosen as an example, input shown in Figure 5A. The FEP simulations now revealed there were no cases of ligands with RMSD exceeding 4 Å. Correspondingly, results again improved, R^2 0.62 and MUE 0.67 ± 0.08 kcal/mol FEP-study 19 Table 2. Importantly the correlation plot and the $\Delta\Delta G$ error count in Figure 5C were substantially improved. Indeed the $\Delta\Delta G$ error count revealed 75% of perturbations were calculated within an accuracy of 1 kcal/mol, which led to a highly accurate set of predicted ΔG over all calculations. These results can be put into the context of the expected R^2 for the 24 molecules based on the experimental or FEP errors, 0.83 and 0.29 respectively, showing that the R^2 from the FEP calculations did not outperform experiment but performed better than a 1 log unit error. Finally, the MM-GBSA calculations did not display any predictive capability with low R^2 of 0.10 and large MUE of 5.01 ± 0.83 kcal/mol seen for the default protocol.

The SAR on the central aryl ring was now captured by the FEP protocol. First, the ring is linked to the pyrrolopyrrole scaffold by an amide in which the carbonyl oxygen forms an important hydrogen bond to a conserved water above that in turn bridges to H350^{7.39}. Thus, the position and conformation of the carbonyl is crucial, in turn, the central aryl ring must twist orthogonal to the amide such that it occupies the subpocket making an offset edge to face interaction with Y354^{7.43}, a similar interaction with H350^{7.39} and also forming the intramolecular interaction to the distal pyrimidine, Figure 5A. The twisted orientation with respect to the amide explains

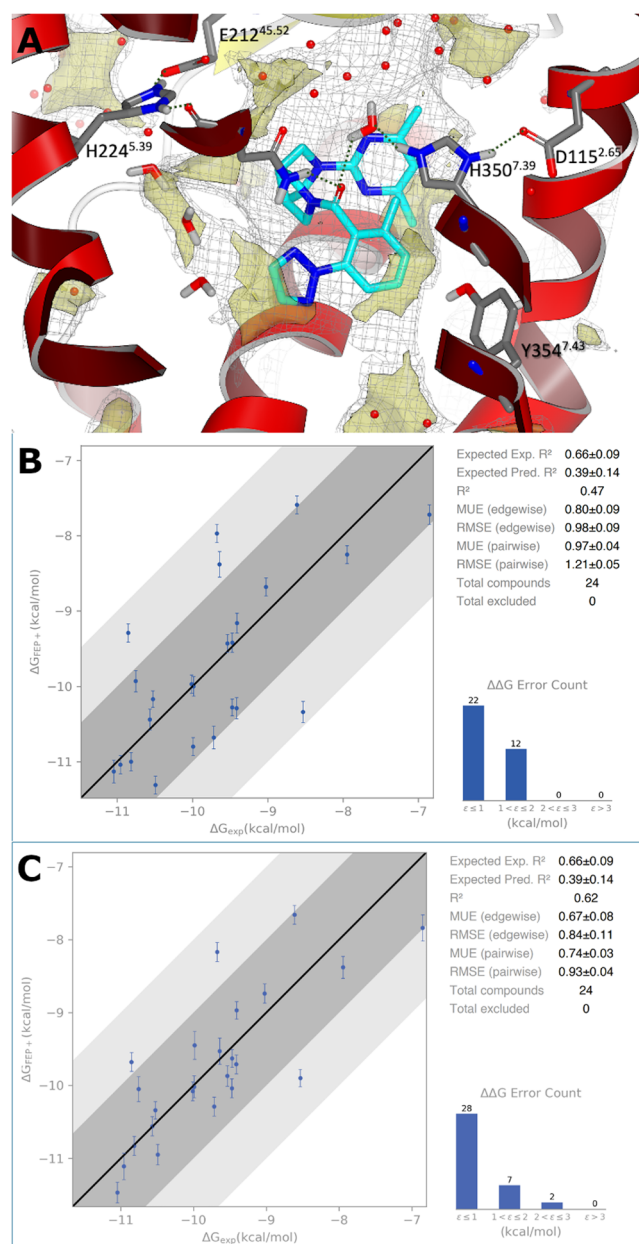


Figure 5. OX2 binding site and FEP results for data set 2. (A) Input configuration for the new input built with pre-equilibration in the presence of the most active data set 2 compound, FEP-study 19. As in Figure 4, the input structure highlights the lipophilic hotspots in the binding site, rendering and color coding of binding site surface (gray mesh), and lipophilic regions (solid yellow) using GRID are as defined for Figure 2. The FEP results are shown with the correlation of calculated vs experimental ΔG (kcal/mol) for FEP-study 18 (B) and FEP-study 19 (C).

the preference for double ortho substitution, triazine on the one side and small substituents such as methyl and fluoro on the other. Meanwhile the lipophilic hotspots shown in Figure 5A reveal that only small substitution is possible on the additional positions of the aryl ring, which is confirmed in the SAR with small groups such as methyl and fluoro being preferred to larger methoxy substituents. Overall, the SAR on this ring is driven by maintaining the interaction between carbonyl and water and fixing an optimal ligand conformation to interact with surrounding aromatic side chains and direct the triazine deeper into the pocket. In summary, with this small

modification to the FEP protocol excellent results were achieved that would be robust for prospective drug discovery application.

■ DISCUSSION AND CONCLUSIONS

RBFE calculations on GPCRs present additional problems to standard applications on soluble globular proteins. Similar difficult modeling issues can arise for other drug target families, such as ion-channels and transporters. The setup of the system is more complex due to the incorporation of the lipid membrane, larger solvent box and deep pockets where waters are trapped by ligands. Our results have shown sensitivity to extended equilibration, equilibration with alternative ligands, or longer simulation times, all suggesting convergence of sampling can be an issue for these systems. In addition, it is critical to be able to observe change in trapped water networks during simulations. This is not particularly surprising as experience suggests that even classical MD simulations on GPCRs requires more careful attention to system setup, equilibration, membrane behavior, etc., to ensure successful and stable outcomes. Our results have confirmed this is also the case in the realm of FEP simulations.

Beyond the large-scale issues of system setup and construction, GPCRs also present challenges due to the greater uncertainty in the small-scale details around the ligand binding site arising from two reasons. On the one hand, not having access to high-quality X-ray (cryo-EM) structures requires a need to rely upon homology models, modeled binding modes, or other modeling assumptions to compensate for low resolution structures. In this regard we consider the uncertainties may be similar to running FEP on any homology model.³⁹ On the other hand, GPCRs have characteristic structure-based design elements⁴⁸ arising from the narrow, deep, and lipophilic binding sites that are not straightforward for computational modeling.

These elements include buried waters that are particularly common in GPCR binding sites and can play a major role in SAR. Sharp changes in ligand activity known as activity cliffs can arise when apparent small ligand modifications lead to displacement of energetically unstable waters resulting in substantial gains in ligand affinity.⁴⁷ The MM-GBSA method used here performed poorly because the continuum solvation approach does not capture these effects. Although some reports using similar end-point FE methods have shown improvements with the inclusion of individual water molecules,^{79,80} arguably such an approach is inappropriate when alternative alchemical RBFE methods using explicit solvent are feasible. Nevertheless, the (dis)placement of buried waters still presents a challenge for modeling with MD simulations due to their kinetic stability, hence methods based on GCMC algorithms have been introduced and shown here to be beneficial for SAR prediction by FEP. In addition, the use of inhomogeneous solvation theory (IST) to identify stable water locations in the binding site can improve RMSD stability and FEP performance. We used the WaterMap software but similar theory can be implemented with tools such as Grid-IST (GIST).⁸¹ Our results have shown the importance of aromatic substituents for affinity in the A_{2A} series, meanwhile for OX2 buried waters in the deepest region of the binding cavity formed the limit or lower boundary of the site and helped stabilize the ligands in their binding mode.

Analysis of GPCR binding sites has revealed a second factor that is the tight complementarity between the binding site,

water structure, and ligand placement. It has been shown that ligand-based models assuming superposition of pharmacophoric features would not be sufficient to recapitulate GPCR ligand SAR and even small differences in ligand overlap can block or allow apparently shared substitution vectors between similar ligands.⁴⁵ Therefore, the need for careful refinement of the ligand position, side-chain orientations, in addition to water placement as mentioned above can be appreciated. Apparently small errors or oversights in system setup could result in unstable ligand RMSD during simulations and poorly performing RBFE calculations.

The GRID analyses of the A_{2A} and OX2 structures have shown how relatively small lipophilic hotspots, often at the bottom of subpockets, are key for SAR. Only when ligands are stable in simulations, and the correct substituents can occupy these regions will FEP be capable of accurately predicting the true changes in affinity. Lipophilic hotspots have previously been shown to be a critical determinant of not only orexin receptor–ligand binding⁴⁶ but also GPCRs in general.⁷⁵ It has been confirmed here again, especially in the case of OX2, where the SAR was linked with these hotspots. Also interesting in this case has been the ability of RBFE calculations to recognize those trends in SAR and correctly predict the outcomes of the structural modifications, directly in the lipophilic hotspot regions. This further highlights the importance of successful molecular design to predict optimal structural modifications in these regions alongside the classical view of polar interactions. Thus, a key factor for success in this study was to include and perform equilibration with representative initial ligand reference structures to define independent FEP models for different subseries, and especially equilibrating the system with bulky substitutions.

Given the importance of the HIS protonation state for both A_{2A} and OX2 cases, we speculate that constant pH simulations could provide a future solution to this type of uncertainty. These approaches are based on alchemical methodology such as λ -dynamics and are capable of considering the titratable states of different amino acids and assessing their likelihood of occurrence from the simulations in an objective manner. Modern implementations are compatible with explicit solvent and reveal good results in relatively fast GPU implementations.^{82,83}

We have previously shown that FEP can deliver accurate results with opensource software,^{28,43} but it is often the case that each system requires bespoke and time-consuming treatment.²⁸ Here we have described the challenging aspects of GPCR RBFE calculations, and how the many different considerations require an FEP workflow with access to varied tools: IST, GCMC water placement, extended sampling, extended equilibration, replica exchange, tools to efficiently analyze trajectories, check convergence, etc. Our work was enabled by the use of the commercial Schrödinger FEP+ approach, however, the underlying theory and tools that we have described are in some cases available in other simulation packages.^{43,84,85} To date, it remains complex to combine all these aspects with most software, but we expect to see further advances in this regard.

In summary, the careful consideration of water placement, ligand stability, equilibration, and sampling can greatly improve FEP results. This study was based on different subseries of ligands for two GPCRs and highlights the common issues. It is not expected that a single protocol or model will work for all GPCR ligand combinations, but the steps taken in this study

can be followed and expected to have a beneficial impact on performance. Our experience in modeling the SAR of GPCRs suggests these themes recur frequently. While high resolution experimental structures can overcome uncertainties of water or side chain position for some targets, the methodological issues such as kinetically trapped waters, or simulation stability remain, and will be relevant for other target classes. We have shown that when these aspects are considered in detail, with sufficient iterative testing and analysis, RBE calculation results for GPCR systems can be achieved with similar accuracy as that reported for nonmembrane proteins.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.0c00449>.

Correlation of calculated and experimental activities for additional studies and the calculated free energy differences for specific perturbation networks (PDF)

Input PDB and ligand structures (ZIP)

■ AUTHOR INFORMATION

Corresponding Authors

Chris de Graaf – Sosei Heptares, Great Abington, Cambridge CB21 6DG, United Kingdom; orcid.org/0000-0002-1226-2150; Phone: +44 (0)1223 949210; Email: chris.degraaf@soseiheptares.com

Gary Tresadern – Computational Chemistry, Janssen Research & Development, Janssen Pharmaceutica N. V., B-2340 Beerse, Belgium; orcid.org/0000-0002-4801-1644; Phone: +32 1464 1569; Email: gtresade@its.jnj.com

Authors

Francesca Deflorian – Sosei Heptares, Great Abington, Cambridge CB21 6DG, United Kingdom; orcid.org/0000-0002-8904-4645

Laura Perez-Benito – Computational Chemistry, Janssen Research & Development, Janssen Pharmaceutica N. V., B-2340 Beerse, Belgium; orcid.org/0000-0001-9607-9048

Eelke B Lenselink – Division of Medicinal Chemistry, Leiden Academic Centre for Drug Research, Leiden University, Leiden 2300, RA, The Netherlands; orcid.org/0000-0001-5459-2978

Miles Congreve – Sosei Heptares, Great Abington, Cambridge CB21 6DG, United Kingdom; orcid.org/0000-0003-1913-0318

Herman W. T. van Vlijmen – Computational Chemistry, Janssen Research & Development, Janssen Pharmaceutica N. V., B-2340 Beerse, Belgium; orcid.org/0000-0002-1915-3141

Jonathan S. Mason – Sosei Heptares, Great Abington, Cambridge CB21 6DG, United Kingdom

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.jcim.0c00449>

Author Contributions

The project was conceived by H.W.T.v.V., J.S.M., C.d.G., and G.T. Experiments were performed and analyzed by F.D., J.S.M., L.P.-B., E.B.L., and G.T. The manuscript written by F.D., J.S.M., C.d.G., and G.T. The manuscript was revised and approved by all authors.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Davide Branduardi, Katalin Phimister, and Thomas Steinbrecher at Schrödinger are acknowledged for technical support and for providing the FEP protocol with extended equilibration step used on the A_{2A} pyridine-triazines FEP-study 9. This project was partly funded by Vlaams Agentschap Innoveren & Ondernemen Project 155028: ACCIDD—Advanced computational chemistry in drug discovery. F.D., L.P.-B., E.B.L., M.C., H.W.T.v.V., J.S.M., C.d.G., and G.T. participate in the European Cooperation in Science and Technology Action CA 18133 [European Research Network on Signal Transduction (ERNEST)].

■ ABBREVIATIONS

FEP, free energy perturbation; GCMC, grand canonical Monte Carlo; GPCR, G protein-coupled receptor; MD, molecular dynamics; MUE, mean unsigned error; ns, nanoseconds; OX2, orexin-2; PDB, Protein Data Bank; RMSD, root-mean-square deviation; RMSE, root-mean-square error; SAR, structure activity relationship

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